

Experimental comparison

Native **DS-TWR** vs. **FlexTDOA**

Field data, surveyed 3×3 m geometry, static centered tag

Test date: July 26, 2026, 20:04–20:18 EEST
Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi 5
Firmware: commit a97cdce60d19, -dirty state
Branch: experiment/ds-twr-native-performance
Configuration: M1 tag; M2, M3, M4, M5 anchors
Environment: LOS, static position

Key result. At this central test point and under the tested conditions, DS-TWR delivered the best unfiltered accuracy: a 2D RMSE of **3.25 cm** and P95 of **4.80 cm**, at **15.15 positions/s**. FlexTDOA delivered much higher radio throughput with a passive tag: **804.6 observations/s**; the embedded solver published **25.20 positions/s**, with a **3.59 cm** median, **7.56 cm** P95, and **4.44 cm** RMSE after the ground-truth-independent structural validity checks described in Section 4.3.

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1 Executive summary

The test addressed two distinct questions:

1. how accurate the measurements and resulting position are without temporal filtering, in the same physical geometry;
2. how quickly and robustly each implementation delivers useful data in the current configuration.

Six alternating 120 s blocks were collected: **Flex1-DS1-Flex2-DS2-Flex3-DS3**. Each protocol therefore has 360 s of observation time, after 15 s of stabilization following each mode change. No dashboard polling errors occurred. The final data set contains 22,256 DS-TWR ranges, 289,716 captured FlexTDOA observations, 5,455 complete DS positions, 50,879 offline FlexTDOA solutions from complete valid slots, and 9,075 positions published by the ESP32 solver. Four FlexTDOA observations and one embedded position failed structural validity checks and remain available in the audit data, but are excluded from all comparative metrics.

Table 1: Aggregate position results after structural validity checks. No position was selected or rejected using its ground-truth error.

Series	N	Hz	Median	P95	RMSE	Maximum
DS-TWR, surveyed geometry	5,455	15.15	3.04 cm	4.80 cm	3.25 cm	6.79 cm
FlexTDOA, offline complete slot	50,879	141.30	7.64 cm	17.17 cm	9.94 cm	84.53 cm
FlexTDOA, ESP32 solver	9,074	25.20	3.59 cm	7.56 cm	4.44 cm	25.44 cm

Preliminary interpretation. At this single static point, DS-TWR produced lower position errors. FlexTDOA provided a passive tag and higher update rate, at the cost of greater dispersion. The result does not establish the general superiority of either protocol: the central LOS point is geometrically favorable, and the test does not cover edges, motion, antenna orientation, or NLOS conditions.

2 Technical background

2.1 Native DS-TWR

The tested implementation uses exactly the three required UWB frames: **POLL**, **RESP**, and **FINAL**. The tag initiates the exchange with each anchor; **FINAL** carries the tag timestamps, and the anchor combines them with its own hardware timestamps to calculate the propagation time. This is the flow described in the official Qorvo DW3xxx examples [2].

For asymmetric DS-TWR, propagation time is estimated using the relationship [3]:

$$\hat{T}_{\text{prop}} = \frac{R_a R_b - D_a D_b}{R_a + D_a + R_b + D_b}, \quad (1)$$

where R_a, R_b are the round-trip intervals and D_a, D_b are the response delays measured by the two clocks. The distance is $\hat{d} = c\hat{T}_{\text{prop}}$.

In the firmware, the timestamps include calibrated antenna delays, and the estimate also uses instantaneous CIA/carrier-integrator correction. These are physical measurement corrections, not temporal filters. Each result is a single **POLL-RESP-FINAL** measurement; the dashboard applies no median, moving average, Kalman filter, or smoothing before trilateration.

2.2 FlexTDOA

FlexTDOA is distributed DL-TDOA: within a slot, one anchor transmits **REQ**, and K anchors respond in sequence. The tag only listens. The initiator and responder order rotate, eliminating a permanently designated reference anchor while keeping the tag radio-passive. The architecture, CI-CR scheduling, and AlgMin come from Pătru et al. [1].

For an observation with initiator i and responder j , the firmware forms:

$$z_{ij} = c \left[(t_j^{\text{tag}} - t_i^{\text{tag}}) - \hat{T}_{\text{reply},j} \right] - \hat{d}_{ij}, \quad (2)$$

and ideally:

$$z_{ij} \simeq \|\mathbf{p} - \mathbf{a}_j\| - \|\mathbf{p} - \mathbf{a}_i\|. \quad (3)$$

The reply term is corrected using instantaneous CFO, and $\hat{d}_{ij}$ is the anchor-to-anchor distance carried in the protocol stream. The AlgMin position minimizes:

$$\min_{\mathbf{p}} \sum_{(i,j)} [\|\mathbf{p} - \mathbf{a}_j\| - \|\mathbf{p} - \mathbf{a}_i\| - z_{ij}]^2. \quad (4)$$

The firmware applies no median, moving average, or Kalman filtering to the observations. The embedded solver uses the most recent still-valid directed observations (at most 12 for four anchors) in a damped least-squares solution. Reusing the latest coherent set is not a temporal average, but it makes the embedded output different from the offline “one slot, three observations” solution reported separately here.

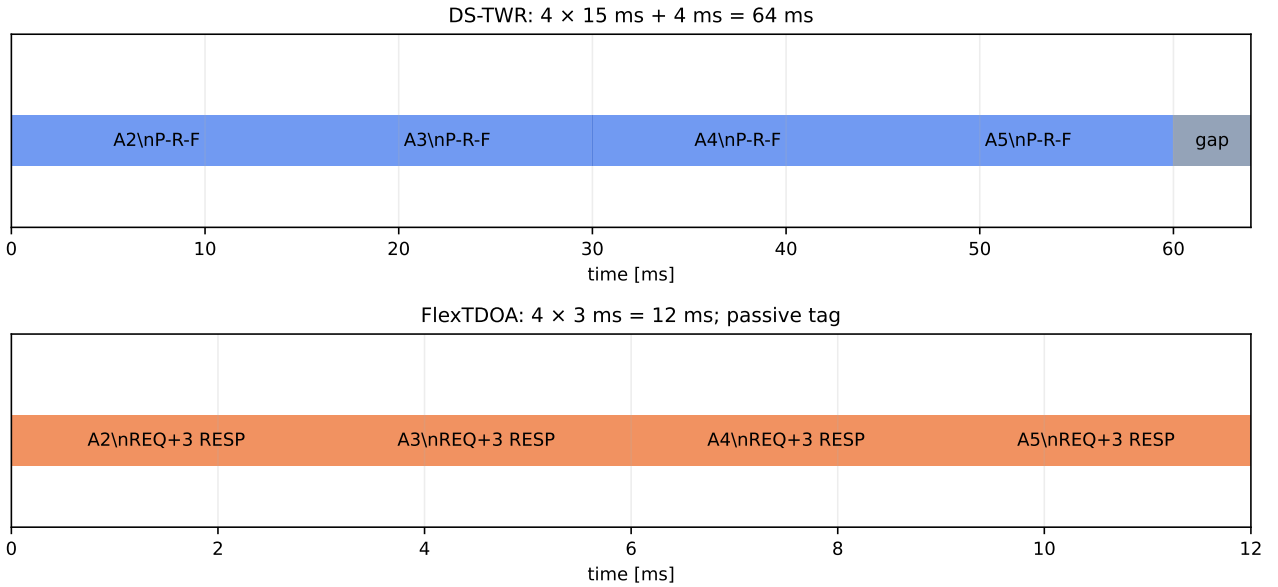


Figure 1: Timing schedule active during the test. P-R-F = POLL-RESP-FINAL.

Table 2: Structural differences relevant to the system.

Characteristic	Native DS-TWR	FlexTDOA
Tag role	Active; transmits two of the three frames for every range	Radio-passive; receive only
Observable	Absolute tag-to-anchor distance	Range difference between two anchors
Frames	3 per range: POLL, RESP, FINAL	1 REQ + up to 3 RESP per slot
Scaling with tags	UWB traffic grows with the number of tags	Additional tags require no additional UWB slots
Solver in test	Linear trilateration on the Raspberry Pi	2D AlgMin on ESP32 plus offline reproduction
Temporal filtering	No	No
Active profile	64 ms/frame with 4 anchors	12 ms/frame with 4 slots

3 Test configuration

3.1 Geometry and reference

The anchors were surveyed as a square with a 3.000 m side and a stated maximum positioning deviation of ± 2 mm. The convention used in the common analysis is:

$$A2 = (0, 0), \quad A3 = (0, 3), \quad A4 = (3, 0), \quad A5 = (3, 3) \text{ m},$$

$$\mathbf{p}_{\text{tag}} = (1.5, 1.5) \text{ m}.$$

The true tag-to-anchor distance is the same for all anchors:

$$d_{\text{true}} = \sqrt{1.5^2 + 1.5^2} = 2.121320344 \text{ m}.$$

At the center, all true TDOA differences are zero. This property allows each z_{ij} to be evaluated directly without introducing solver error into the measurement analysis.

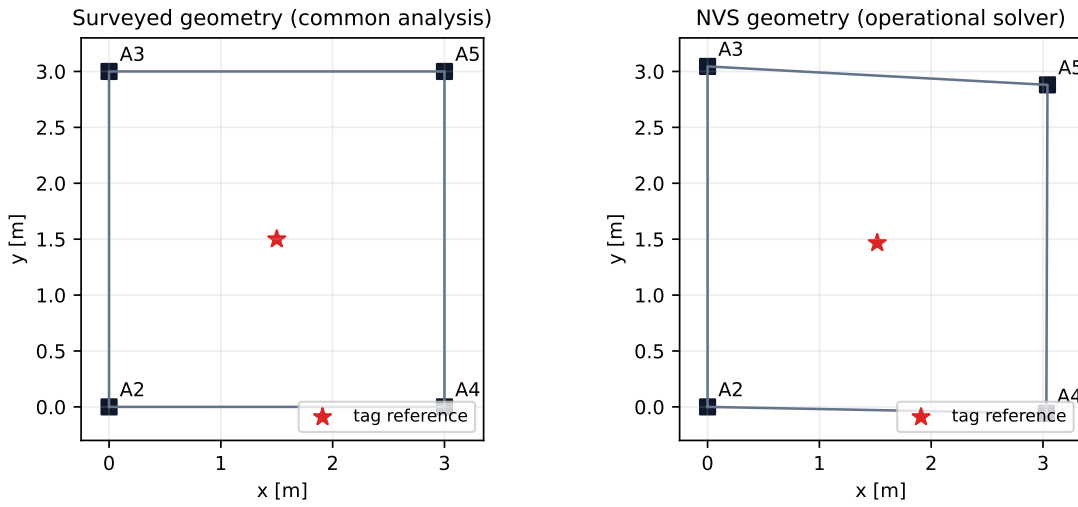


Figure 2: Surveyed geometry used for the common comparison and NVS geometry used by the operational solver.

The fixed geometry stored in NVS at the time of the test was:

Anchor	NVS coordinate [m]	Surveyed coordinate [m]	Difference [mm]
A2	(0.000, 0.000)	(0.000, 0.000)	(0, 0)
A3	(0.000, 3.045)	(0.000, 3.000)	(0, +45)
A4	(3.030, -0.058)	(3.000, 0.000)	(+30, -58)
A5	(3.040, 2.880)	(3.000, 3.000)	(+40, -120)

Consequently, the embedded FlexTDOA positions were evaluated in their operational coordinate frame relative to the center of the NVS geometry, (1.5175, 1.46675) m. For the direct protocol comparison, both offline solutions use the surveyed square and the (1.5, 1.5) m reference. Keeping these evaluations separate prevents anchor self-localization error from being attributed to the positioning protocol.

3.2 Radio and timing parameters

Both protocols used the same DW3000 hardware, channel 5, 6.8 Mb/s, 64 MHz PRF, and preamble length 128. The DW3000 family is designed for TWR and TDoA, and the manufacturer specifies ranging accuracy below 10 cm in its product specification [4, 5]; the result in this report, however, applies only to the specific platform and environment tested.

Table 3: Active runtime profiles read from all five modules.

Parameter	Value	Interpretation
DS slot	15 ms	budget for one native exchange
DS gap after 4 anchors	4 ms	complete frame $4 \times 15 + 4 = 64$ ms
DS RX slice / timeout	100 / 8 ms	anchor listening / expected-frame timeout
DS RESP / FINAL delay	2 / 2 ms	hardware delayed TX
DS auto-RX delay	500 UUS	RX opening after TX
Flex guard	250 μ s	per-slot guard
Flex REQ subslot / process	250 / 1,000 μ s	request and processing
Flex RESP subslot	250 μ s	spacing between responses
Flex process per responder	250 μ s	3×250 μ s
Flex slot / frame	3 / 12 ms	333.33 slots/s; 83.33 frames/s

The active antenna delays, in DW3000 register units, were: M1 16,363, M2 16,368, M3 16,362, M4 16,371, and M5 16,371.

4 Methodology

4.1 Experimental plan

- six alternating 120 s Flex–DS blocks, reducing confounding between the protocol and thermal or temporal drift;
- a 15 s warm-up after every mode change;
- dashboard polling at 50 Hz and deduplication by event ID, sequence, slot, and pair;
- full status at 1 Hz; all five modules remained online and in the expected mode;
- zero polling errors across all six blocks;
- the system was left in DS-TWR mode at the end.

4.2 Three position evaluations

Offline DS-TWR, surveyed geometry. A position is calculated only when one range for each of A2–A5 exists within the same frame. The same unweighted linear trilateration as in the dashboard was used. No ranges were removed based on error magnitude.

Offline FlexTDOA, surveyed geometry. A slot is solved only when the tag received all three responders. The 2D AlgMin implementation reproduces the dashboard solver convention and iterations, starting deterministically at the center of the square. Each solution uses only that slot; there is no temporal seed, averaging, or filtering.

Embedded FlexTDOA, operational geometry. The positions published directly by the ESP32-S3 are analyzed. They use up to 12 recent directed differences and the NVS geometry. This is the series relevant to what the user sees in the dashboard.

4.3 Structural validity rules and metrics

The comparison uses validity checks that do not depend on the known tag position. A corrected FlexTDOA observation is accepted only if it satisfies the universal range-difference bound

$$|z_{ij}| \leq \|\mathbf{a}_i - \mathbf{a}_j\| + 0.01 \text{ m.}$$

Four of 289,716 captured observations failed this check. An embedded position is accepted only when the solver’s own `rms_m` does not exceed the anchor-array diagonal, $\sqrt{3^2 + 3^2} = 4.243$ m. One of 9,075 published positions failed this deliberately permissive self-consistency check. DS-TWR had no corresponding structural failure. All excluded records remain in the derived CSV files with a validity flag and in the immutable raw captures.

No ground-truth error threshold, percentile trimming, or manual event ID is used for selection. The same accepted data population is used consistently for aggregate metrics, per-block metrics, and plots.

We report axis bias, median radial error, P95, P99, maximum, 2D RMSE, and precision relative to the sample centroid (CEP50/CEP95). For measurements, we report bias, standard deviation, MAE, absolute P95, and robust deviation $1.4826 \times \text{MAD}$.

Repeatability is evaluated across the three blocks rather than treating hundreds of thousands of temporally correlated samples as independent observations. Where a 95% uncertainty is stated, it is the Student interval of the mean of the three blocks ($t_{0.975,2} = 4.303$). With only three blocks, the interval should be interpreted descriptively.

5 Results

5.1 Throughput, availability, and position rate

Table 4: Per-block results. For Flex, “complete” means three distinct responses in a slot; for DS, four ranges in a position frame. The Flex position is the embedded output. Flex counts and rates shown here are after structural validation; 0, 2, and 2 observations and 0, 1, and 0 positions were excluded from Flex 1, 2, and 3, respectively.

Block	Measurements	Measurements/s	Theoretical delivery	Complete	Positions/s	Median / P95
Flex 1	95,932	799.07	79.91%	40.82%	25.66	3.72 / 7.56 cm
DS 1	7,426	61.88	99.01%	97.70%	15.21	3.05 / 4.84 cm
Flex 2	96,808	806.69	80.67%	43.01%	24.94	3.56 / 7.57 cm
DS 2	7,400	61.66	98.66%	96.41%	15.01	3.05 / 4.83 cm
Flex 3	96,972	807.95	80.80%	43.45%	25.00	3.49 / 7.52 cm
DS 3	7,430	61.91	99.06%	97.91%	15.24	3.01 / 4.73 cm

The block-level means were:

- DS-TWR: 61.82 ± 0.34 ranges/s and 15.15 ± 0.31 complete frames/s (95%);
- FlexTDOA: 804.57 ± 11.93 valid observations/s, 141.30 ± 11.68 solvable complete slots/s, and 25.20 ± 1.00 valid embedded positions/s (95%);
- the Flex observation rate was 13.0 times the DS range rate; the complete Flex slot rate was 9.3 times the complete DS frame rate; the actual embedded output was 1.66 times faster than the DS position rate.

DS delivery was 98.91% of the profile maximum, and 97.34% of reconstructed candidate frames contained all four ranges. With Flex, the tag observed 80.46% of the theoretical maximum of 1,000 differences/s; 42.43% of slots contained all three responders. A rolling solver can nevertheless publish positions using recent pairs, so the percentage of complete slots is not identical to embedded-position availability.

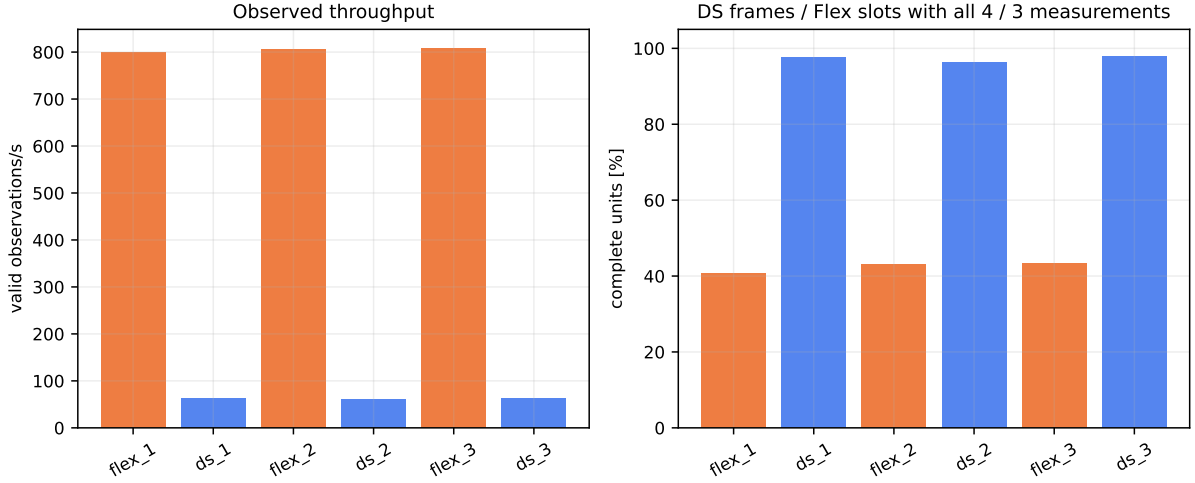


Figure 3: Observed throughput and completeness of position units. The throughput axes compare different observables: DS absolute ranges and Flex range differences.

5.2 DS-TWR measurement quality

Table 5: DS-TWR range error relative to 2.121320344 m.

Anchor	N	Bias	σ	σ_{MAD}	MAE	Absolute P95
A2	5,561	+2.45 cm	1.77 cm	1.78 cm	2.56 cm	5.37 cm
A3	5,555	+8.02 cm	0.92 cm	0.89 cm	8.02 cm	9.77 cm
A4	5,571	+3.03 cm	1.53 cm	1.63 cm	3.04 cm	5.47 cm
A5	5,569	-4.21 cm	1.56 cm	1.48 cm	4.22 cm	6.53 cm

DS-TWR has low dispersion on each link, but its bias is module-dependent: from -4.21 cm at A5 to $+8.02$ cm at A3. This indicates a clear opportunity for per-module/antenna recalibration. The opposing biases project partly favorably onto the central position; the position result should not be interpreted as evidence that every range is calibrated to within 3 cm.

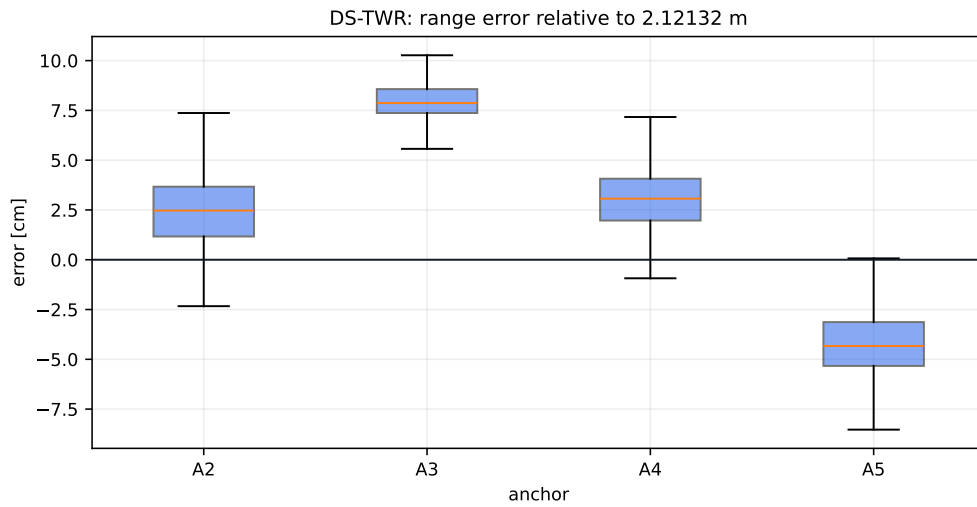


Figure 4: Unfiltered distribution of DS-TWR errors. Outliers are not drawn in the boxplot, but they are included in every metric. The maximum absolute error was 11.47 cm.

5.3 FlexTDOA observation quality

At the center, the true value for every directed pair is zero. The CFO plus anchor-to-anchor propagation correction is essential:

Table 6: The 289,712 structurally valid FlexTDOA observations, aggregated. The nearly zero corrected global bias includes cancellation among directed biases and should not be interpreted on its own.

Series	Bias	σ_{MAD}	MAE	P95 abs.	P99 abs.	Max. abs.
Corrected	0.02 cm	12.31 cm	9.94 cm	24.10 cm	36.40 cm	211.4 cm
Uncorrected	71.78 cm	35.43 cm	71.92 cm	122.30 cm	131.40 cm	161.2 cm

Across pairs, corrected bias ranges from -13.61 to $+13.18$ cm, while robust dispersion ranges from 7.71 to 9.34 cm. The correction drastically reduces the normal offset but does not eliminate the distribution tails.

Exactly four corrected observations violated the physical inequality $|z_{ij}| \leq \|\mathbf{a}_i - \mathbf{a}_j\| + 1$ cm: two in Flex 2 and two in Flex 3, all with A3 as initiator, in two isolated slots. The values were approximately 79.6–79.7 m. They are retained in the audit data but are not used by the measurement statistics, offline solver, or comparative plots. The exclusion rate is 4 out of 289,716, or 0.00138%.

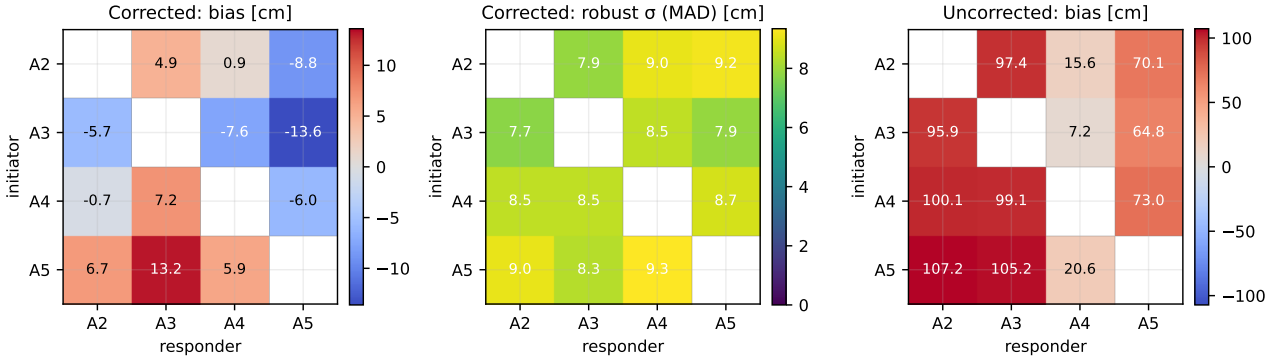


Figure 5: Bias and robust dispersion by directed pair after structural validation. The uncorrected raw field primarily contains the reply delay and clock offset.

5.4 Position accuracy

Table 7: Aggregate position metrics after structural validity checks. CEP is calculated relative to each distribution’s own centroid and describes precision, not absolute accuracy.

Series	x/y bias	RMSE	Median	P95	P99	CEP50/95	> 50 cm
DS, surveyed geometry	+2.63 / -0.99 cm	3.25	3.04	4.80	5.51	1.35 / 2.78 cm	0 / 5,455
Flex, offline complete slot	+4.53 / $+0.60$ cm	9.94	7.64	17.17	25.50	6.44 / 15.42 cm	53 / 50,879
Flex, operational ESP32	+2.29 / $+0.39$ cm	4.44	3.59	7.56	10.44	2.84 / 6.64 cm	0 / 9,074

DS-TWR was more precise and more accurate in the raw comparison on the surveyed geometry. Relative to DS, FlexTDOA solved from individual slots had 3.06 times the RMSE and 3.58 times the P95, but delivered 9.3 times as many solvable complete slots per second.

The embedded FlexTDOA solver is significantly more stable than a single-slot solution because it uses multiple recent pairs: P95 is 7.56 cm and RMSE is 4.44 cm. P95 is 1.58 times the DS value, while RMSE is 1.37 times the DS value. This improvement is not the result of an explicit temporal filter; it comes from overdetermination and reuse of the current observation set.

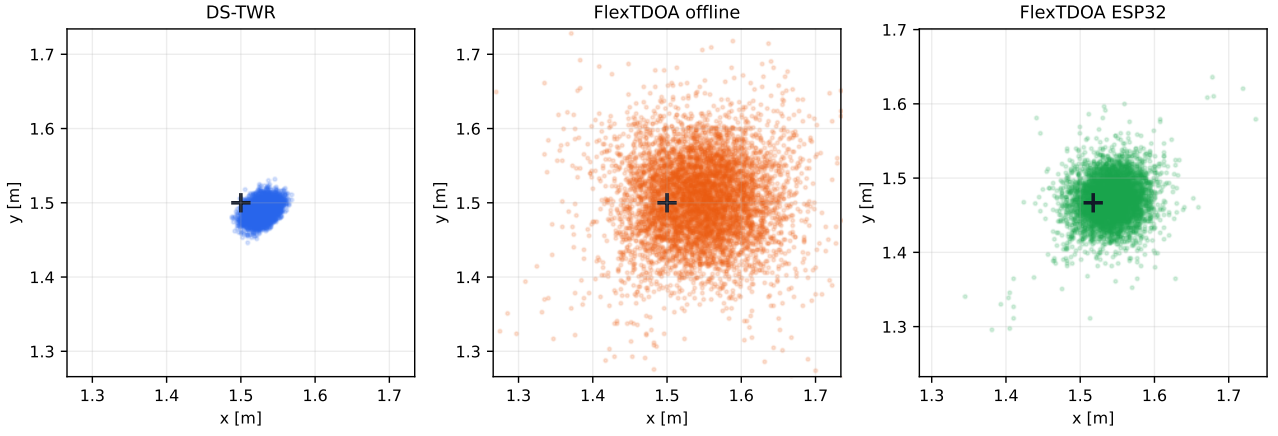


Figure 6: Position clouds. The cross marks the reference for the respective geometric frame. All plotted positions passed structural validation.

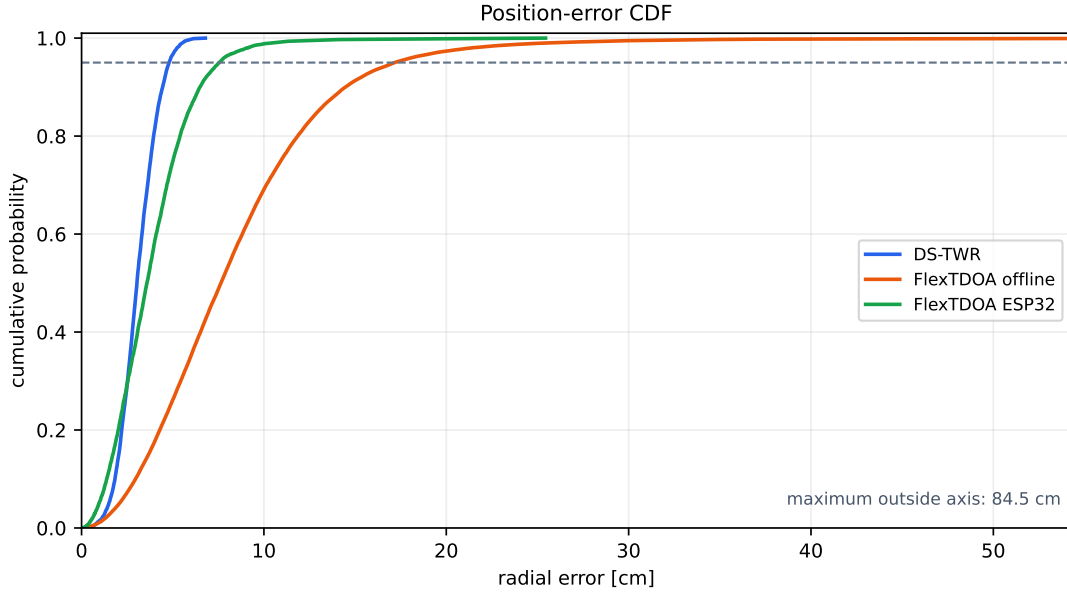


Figure 7: Cumulative distribution of radial error using all samples. The axis limit is robust; the maximum outside the axis is annotated in the plot.

5.5 Records excluded by structural validation

The range-difference bound excluded four FlexTDOA observations, and the solver-self-consistency bound excluded one embedded position. The latter had an internal RMS of 21.854 m, more than five times the permissive 4.243 m limit; the next-highest internal RMS in the data set was 0.617 m. Therefore, the exclusion is determined by a large, isolated break in solver consistency, not by its position error relative to the known center.

The raw captures and the unfiltered embedded-position CSV retain the excluded records and their reasons. This preserves auditability while preventing an invalid protocol state from dominating the static accuracy comparison.

5.6 Temporal repeatability

DS-TWR had per-block RMSE values of 3.20–3.28 cm and P95 values of 4.73–4.84 cm. Mean per-block RMSE is 3.248 ± 0.095 cm (95%). Offline Flex had RMSE of 9.90–10.01 cm and P95 of 16.96–17.34 cm, with a mean RMSE of 9.940 ± 0.154 cm. Embedded Flex had RMSE of 4.39–4.50 cm, with a mean of 4.443 ± 0.127 cm, and P95 of 7.52–7.57 cm.

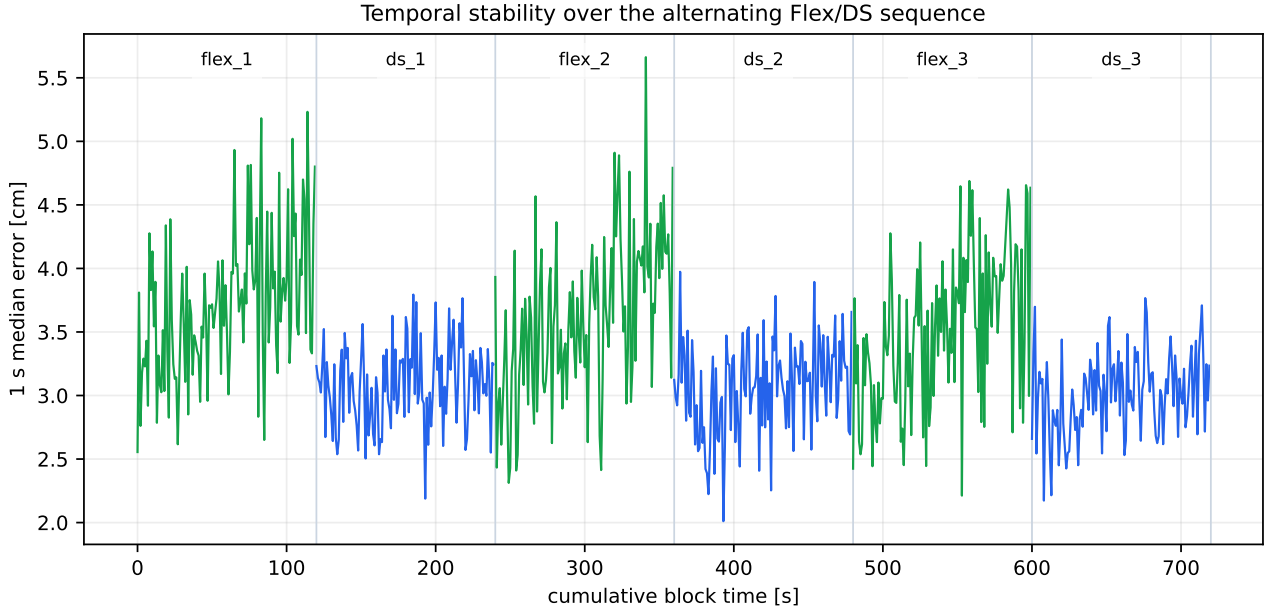


Figure 8: Median error in 1 s intervals. Individual events are not visible in a per-second median and are analyzed separately.



Figure 9: Per-block RMSE and P95 for the operational output: offline DS and embedded Flex, after structural validation.

6 Interpretation and protocol selection

6.1 When DS-TWR is the appropriate choice

- one tag or few tags, where the additional traffic is acceptable;
- accuracy and short distribution tails have priority;
- it is acceptable for the tag to transmit and consume radio energy;
- a rate of approximately 15 positions/s meets the application's dynamics.

In the current test, DS has the best combination of RMSE, P95, between-block stability, and frame completeness. Per-anchor bias nevertheless shows that recalibration could improve accuracy further or

alter it at other positions.

6.2 When FlexTDOA is the appropriate choice

- a fully UWB-passive tag, with battery life and scalability as priorities;
- multiple tags that must be located without consuming additional slots;
- a high observation rate and faster updates;
- the application tolerates a P95 of approximately 7.6 cm in the current configuration and validates impossible outputs.

FlexTDOA is not simply “faster DS”; its observable, topology, and scaling cost are different. With four anchors, the 12 ms profile produces 83.33 frames/s and 333.33 slots/s, independent of the number of passive tags. For a multi-tag installation, the airtime comparison becomes progressively more favorable to FlexTDOA.

6.3 Immediate recommendations

1. Retain unfiltered native DS-TWR with the 64 ms profile as the accuracy baseline.
2. Recalibrate the antenna delays, especially for the A3 and A5 links, and then repeat the test exactly.
3. Write the surveyed geometry to NVS before operational comparisons; the current geometry differs by up to 12 cm on the y axis.
4. For FlexTDOA, add the physical gate $|z_{ij}| \leq d_{ij} + \varepsilon$ and a quality gate on RMS/bounding box. These are validity checks, not temporal filtering.
5. Repeat the test on a grid of at least 5×5 points, including the edges and outside of the square, followed by dynamic and NLOS testing. The center alone does not characterize GDOP over the full area.
6. For motion, report end-to-end latency using hardware timestamps, not only the rate of events received by the dashboard.

7 Limitations

- A single, static, central point cannot describe spatial error.
- The test was LOS; NLOS and severe multipath are not represented.
- The geometry is stated to ± 2 mm, but no external optical system or total station was included in the digital capture.
- Temperature and RSSI were recorded but not varied in a controlled way.
- The DS position is calculated on the Raspberry Pi, while the operational Flex position is calculated on the ESP32; the report therefore also includes a common offline Flex solution.
- Throughput observed through telemetry/dashboard is an end-to-end implementation limit, not only the protocol’s PHY capacity.
- The data set comes from `-dirty` firmware; the manifest and commit provide identification, but exact reproduction requires preservation of the local changes that were active.

8 Preliminary findings

For the surveyed 3×3 m configuration with a centered tag under LOS:

- **DS-TWR produced lower static position error at this point:** RMSE 3.25 cm, median 3.04 cm, P95 4.80 cm, with no errors above 6.79 cm among complete positions;
- **FlexTDOA provided higher throughput and inherent multi-tag scalability:** 804.6 observations/s, a radio-passive tag, and 25.2 embedded positions/s;
- **Embedded FlexTDOA remained close to DS-TWR in its accepted population,** with a 3.59 cm median, 7.56 cm P95, and 4.44 cm RMSE;
- **the validity policy had negligible data-volume impact:** four of 289,716 Flex observations and one of 9,075 embedded positions were excluded without consulting ground-truth position error;
- **the product determines the choice:** DS for accuracy with few tags; FlexTDOA for passive and numerous tags at a high update rate.

These findings are repeatable for the measured static central point because of the alternating blocks, large data volume, and between-block stability. They are not a general protocol ranking. A spatial LOS/NLOS grid, dynamic trials, and antenna-orientation trials are required before making a system-level choice.

A FlexTDOA metrics by directed pair

Table 8: Structurally valid corrected FlexTDOA observations; the true value is 0 m for the centered tag. The final column records source observations excluded by the physical bound. $\sigma_{\text{MAD}} = 1.4826 \times \text{MAD}$.

Pair	N	Bias [cm]	σ_{MAD} [cm]	P95 abs. [cm]	P99 abs. [cm]	Max. abs. [cm]	Excluded by bound
A2→A3	25,583	+4.92	7.86	20.00	30.60	101.60	0
A2→A4	25,192	+0.89	9.04	20.00	31.21	108.60	0
A2→A5	24,354	−8.80	9.19	26.40	52.75	211.40	0
A3→A2	23,516	−5.71	7.71	21.20	36.20	111.60	1
A3→A4	22,517	−7.64	8.45	23.10	34.20	82.50	1
A3→A5	20,732	−13.61	7.86	27.90	37.80	87.10	2
A4→A2	24,892	−0.69	8.45	18.90	30.80	93.60	0
A4→A3	24,316	+7.18	8.45	23.10	35.49	104.40	0
A4→A5	23,612	−5.98	8.75	22.10	33.30	100.80	0
A5→A2	24,976	+6.73	9.04	24.90	42.20	177.40	0
A5→A3	24,931	+13.18	8.30	28.55	37.77	124.10	0
A5→A4	25,091	+5.87	9.34	23.20	34.50	107.80	0

B Files and reproduction

The artifacts are stored in:

`/home/pi/Documents/UWB/reports/uwb_protocol_comparison_20260726`

File/directory	Contents
<code>data/*.jsonl.xz</code>	lossless raw-capture archives, one file per block
<code>data/*.metadata.json</code>	configuration, status, and counters for each block
<code>raw_data_manifest.csv</code>	size and SHA-256 for every raw and compressed file
<code>analysis_summary.json</code>	all aggregate metrics
<code>measurement_metrics.csv</code>	per-anchor/pair metrics
<code>position_metrics.csv</code>	aggregate position metrics
<code>block_metrics.csv</code>	results for all six blocks
<code>ds_positions.csv</code>	reconstructed DS positions
<code>flex_slot_positions.csv</code>	Flex positions from complete slots
<code>flex_embedded_positions.csv</code>	positions published by the ESP32
<code>figures/</code>	all plots in vector PDF and PNG formats

Reproducing the analysis

```
cd /home/pi/Documents/UWB/reports/uwb_protocol_comparison_20260726
xz --decompress --keep data/*.jsonl.xz
cd /home/pi/Documents/UWB
python3 tools/uwb_compare_analyze.py
cd reports/uwb_protocol_comparison_20260726
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The collection script is `tools/uwb_compare_collect.py`, and the analysis script is `tools/uwb_compare_analyze.py`. The analysis does not modify the raw files.

References

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